

Livestock-associated methicillin-resistant *Staphylococcus aureus* responsible for human colonization and infection in an area of Italy with high density of pig farming

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Background Livestock-Associated MRSA (LA-MRSA) belonging to ST398 lineage, common among pigs and other animals, emerged in Central and Northern Europe, becoming a new risk factor for MRSA among farm workers. Strains belonging to ST398 can be responsible for human colonization and infection, mainly in areas with high livestock-farming. The aim of this study was to investigate the occurrence of livestock-associated methicillin-resistant *Staphylococcus aureus* (LA-MRSA) human colonization and infections in an area of the Lombardy Region (Italy), the Italian region with the highest density of pig farming. **Methods** In the period March-April 2010, 879 nasal swabs were taken from subjects at admission to a local hospital serving an area of the Lombardy Region devoted to agriculture and farming. In the period March 2010-February 2011, all MRSA strains from community-acquired infection (CAI) observed in the same hospital, were collected. Molecular characterization of the isolates included SCCmec typing, spa typing and multilocus sequence typing (MLST). **Results** Out of 879 nasal swabs examined, 9 (1%) yielded MRSA. Five strains were assigned to sequence type (ST)398 (spa t899, 3 isolates; t108 and t2922, 1 isolate each) and were therefore categorized as LA-MRSA. The other 4 isolates were likely of hospital origin. No strains were positive for Panton-Valentine Leukocidin genes. Twenty MRSA isolates were detected from CAI, 17 were from skin and soft-tissue infections and 3 from other infections. An MRSA isolate from otitis externa was t899/ST398 and PVL-negative, hence categorized as LA-MRSA. Four isolates were assigned to t127/ST1. Eight strains were PVL-positive community acquired (CA)-MRSA and belonged to different clones, the most frequent being ST8. **Conclusions** In an area of Italy with high density of pig farming, LA-MRSA is able to colonize the population and rarely to produce infections. Typical CA-MRSA is more common than LA-MRSA among CAI.